

What determines switch versus return transitions during binocular rivalry?

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Binocular Rivalry, Exclusive & Mixed Dominance

When the left and right eyes are presented with incompatible images, perceptual experience alternates between them. In exclusive dominance states, perceptual experience is of an image presented to either the left or the right eye. Exclusive dominance states are bridged by mixed dominance states, where perceptual experience is a patchwork of the two images dynamically changing over time.



Switch & Return Transitions

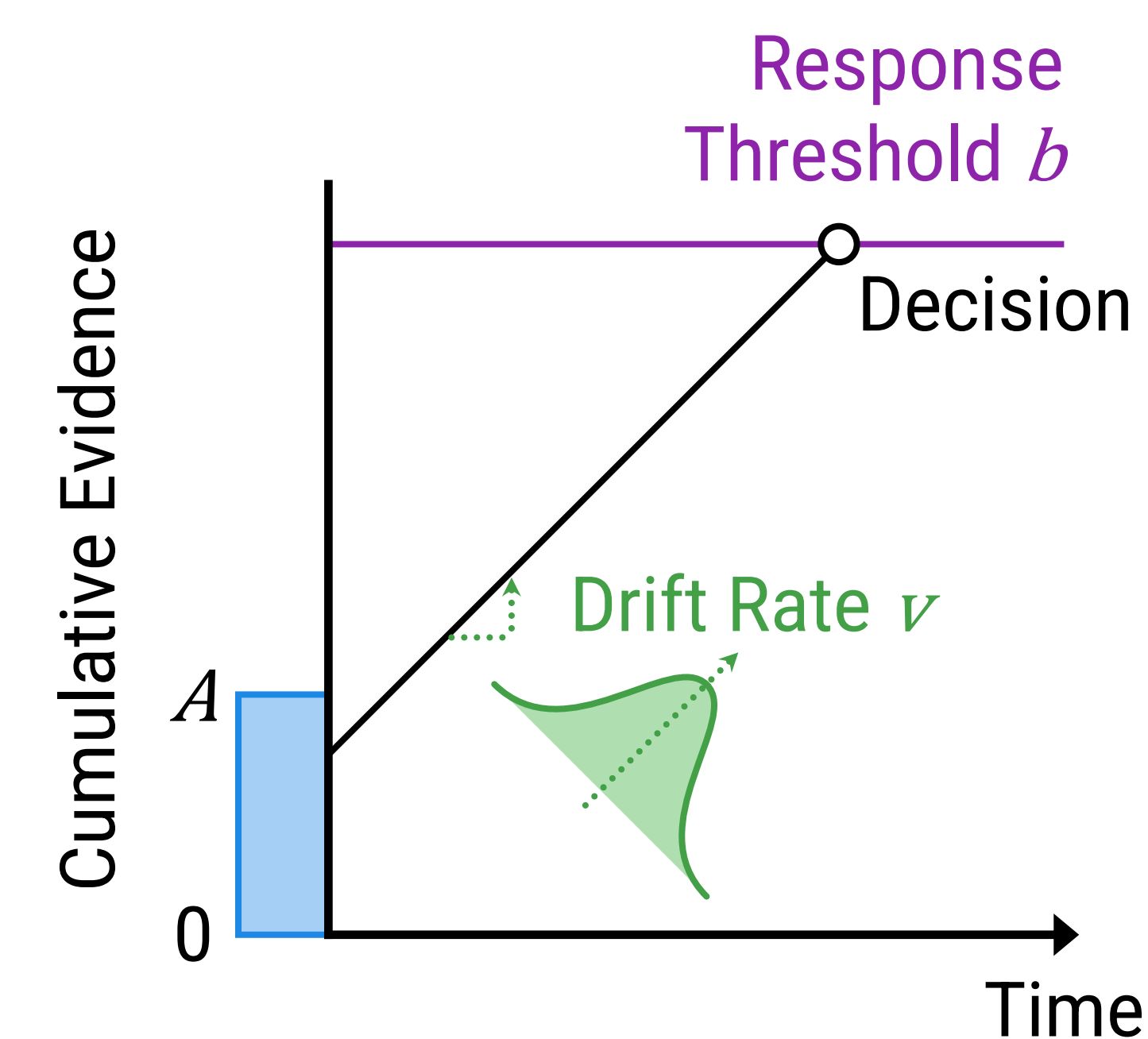
A mixed dominance state can bridge an exclusive dominance state of one eye to that of either the other eye (switch transition) or the same eye (return transition).



Linear Ballistic Accumulator (LBA)

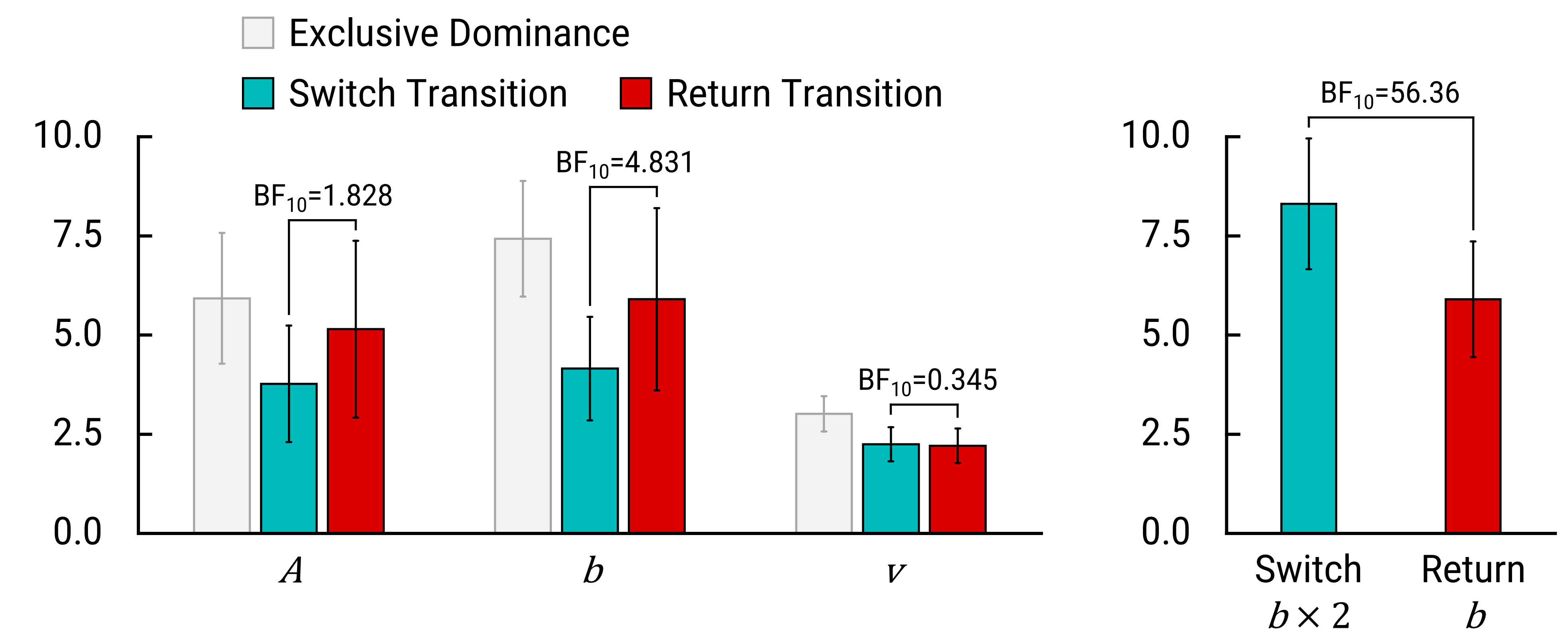
The Linear Ballistic Accumulator (Brown & Heathcote, 2008) models the evidence accumulation process involved in reaching a decision. In each trial of a simulation, the start point of evidence accumulation is sampled from a **uniform distribution ranging between 0 and A** (start point variability parameter), and the slope of evidence accumulation is sampled from a **normal distribution centered around v** (drift rate parameter). The model makes a decision when the cumulative evidence crosses the **response threshold b** .

Separate LBA parameters were estimated for switch and return transition durations, allowing drift rates and response thresholds of the two transition types to be compared.



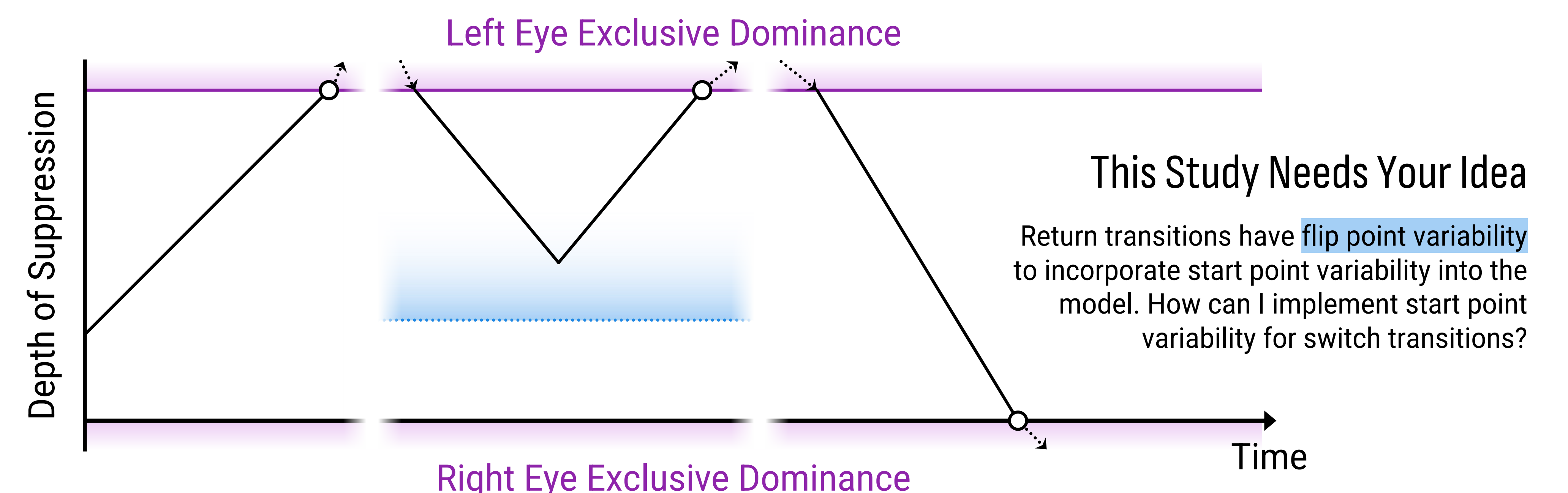
LBA Parameters of Switch vs. Return Transitions

LBA parameters were estimated separately for exclusive dominance, switch transition, and return transition durations sourced from Sy et al. (2016). The drift rates were comparable between the two transition types, and the response threshold for return transitions was larger than that for switch transitions, but did not reach twice the response threshold for switch transitions.



Modeling Exclusive & Mixed Dominance Durations with LBA

The comparable drift rates for switch transitions and return transitions allow for building a unified LBA that models both transition types, by adding a stochastic component to flip the slope of evidence accumulation and using both 0 and b as response thresholds. In this model, the regions outside the response thresholds represent exclusive dominance states, and the y-axis represents the depth of suppression.



References

- Brown, S. D., & Heathcote, A. (2008). The simplest complete model of choice response time: Linear ballistic accumulation. *Cognitive Psychology*, 57(3), 153-178.
- Sy, J. L., Tomarken, A. J., Patel, V., & Blake, R. (2016). The time course of binocular rivalry during the phases of the menstrual cycle. *Journal of Vision*, 16(15), 22.